POLS 641
Introductory Political Analysis
Class 9: Dummy & Categorical Variables

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Northern Illinois University

October 27, 2026

Agenda

- Warm-Up: Final Project Models
- Course Logistics
- R-squared
- Dummy Variables
- Dummy Variables and Experiments
- Categorical Variables

Final Project Check-in

1. Write out a version of your multivariate model

$$Y_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 ..... + \epsilon_i$$

2. Explain model briefly to a new partner - partner point out omitted variable bias
3. Update on data collection and **variable operationalization**
4. Discuss any issues you are struggling with

Course Logistics

Upcoming Schedule

- Today/Thur Dummy and Categorical Variables
- 11-3 Election day no class
- 11-5 Model Specification combined with R
- 11-10/12 Causal Inference & Experiments
- **11-17/19 Review & Midterm**
- 11-24 Thanksgiving Week - No Class (Tuesday - Hackathon?)
- 12-1 Mixed Methods
- 12- 3 Research Presentations
- 12-13 Final Project Due

Coding + Misc Skills

In addition to core topics I plan to discuss:

- More merging + Missing data
- Tidying data
- Principles of data visualization

What other topics would you like to see covered?

- Binary DVs(Logit/Probit/Linear Probability Models)
- Mapping in R
- Factor Analysis
- Power Analysis
- Animation
- What else? You will cover some of these in 642

Upcoming Dates and Deadlines

- 10/28 - Pset # 4 Due
- 11/1 - *Final Project Abstract and Key Descriptive Statistics*
- 11/4 - Datacamp
- 11/11 - Pset # 5 Due
- 12/3 - *Final Project Presentations*
- 12/13 *Final Project due*

Review: Multivariate Regression

Multivariate Regression

What we can interpret:

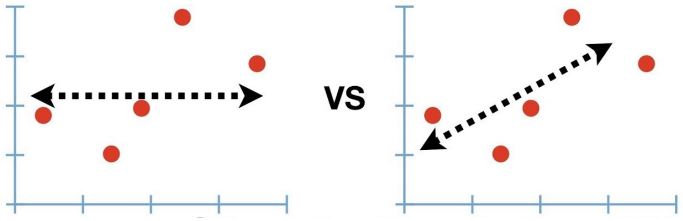
1. Significance - p-value
2. Sign - coefficient positive or negative
3. Size - coefficient's value
4. Substantive - stated in real world terms

What variables?

- Literature
- Theory
- Of course limited to data availability and subject to all caveats on data quality and measurment validity

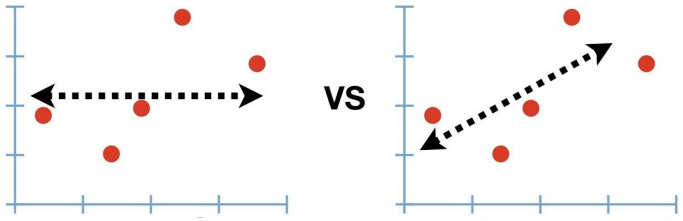
Understanding the "Variance Explained" or R^2

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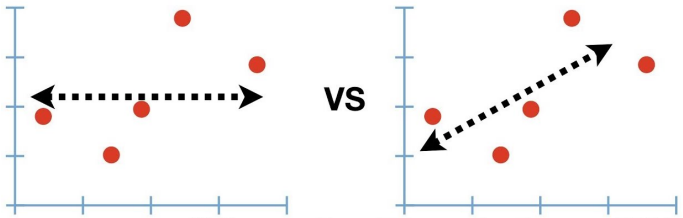
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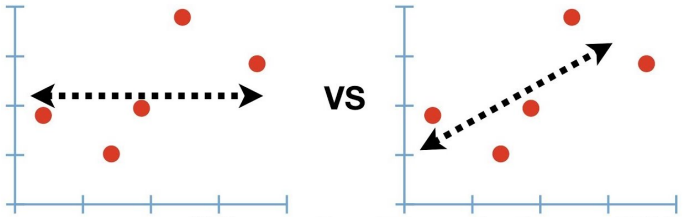
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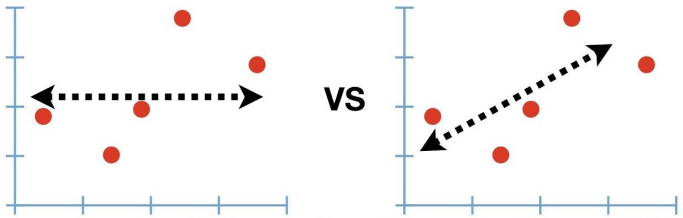
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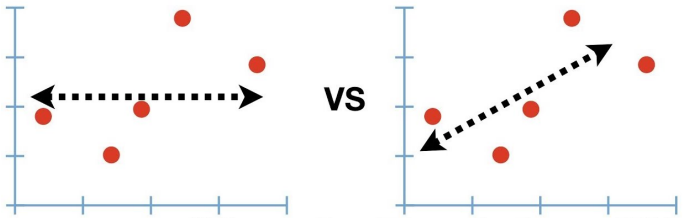
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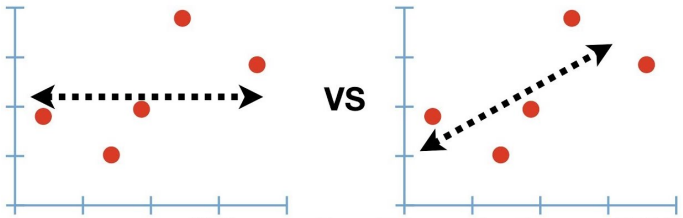
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- this is R^2 statistic at the bottom of the table

- R^2 is interpreted as the proportion of variance explained by the model as whole.
- R^2 is equal to the square of the correlation between the fitted and observed values of the dependent variable.
- R^2 **can also be interpreted as the proportion of variance explained.**
- A highR^2 is neither necessary nor sufficient for an analysis to be useful. A good model can have a low r^2 and a bad model can have a high r^2

- R^2 is our primary means of making comparisons between models

adjusted r^2

- Adding additional variables necessarily increases the r^2
- Adjusted r^2 takes the number of variables into account(using math beyond our scope)
- Interpreted the same way as normal r^2. Usually what is reported when presenting multivariate models.

Review of key points on OLS so far...

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- Including additional ”control" variables decreases omitted variable bias
- Relatedly, we can estimate our model’s ‘goodness of fit’: R^2
- Just because we see a significant relationship in our regression model, doesn’t mean its causal

Review: Interpretation

For Each Independent Variable:

1. Significance
2. Sign
3. Size
4. Substantive

(Which variables and to what extent we interpret them is dependent on our goals for the paper) For the model as a whole:

1. R^2 Proportion of variance explained

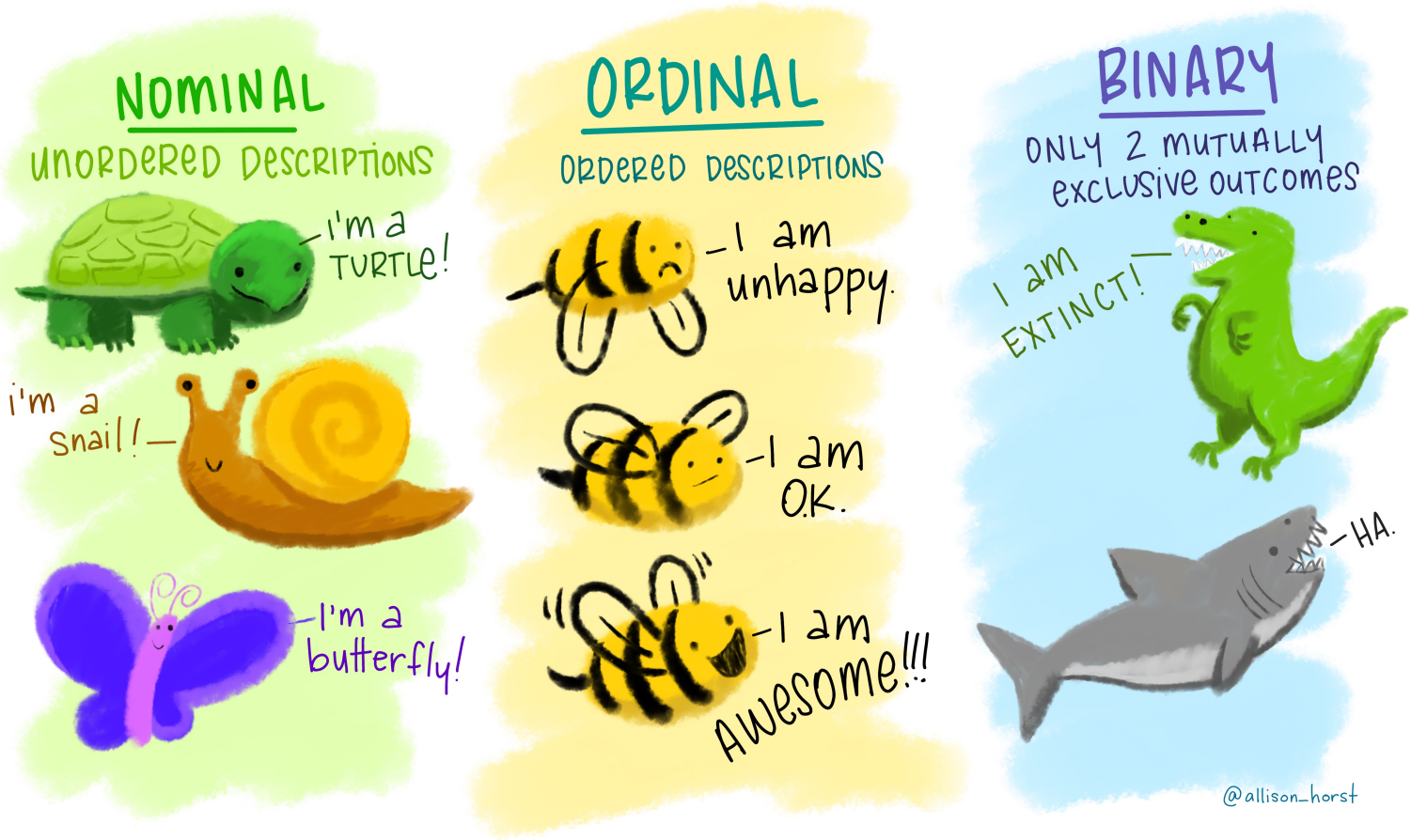
Terminology: Types of Variables



Amazing art from @allison_horst

Terminology: Types of Variables

1. Nominal/Categorical: Discrete groups without an order
   Factor or String data structure in R
2. Ordinal: Ordered but without consistent distance
3. Continuous/interval: Ordered and consistent distance
   Numeric or integar data structure in R
4. Ratio



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Types of Variables & Data

Variables in a data set come in different forms:

- Continuous
  - Transformed variables (like logs, see Ch. 7 and will discuss next week)

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- Continuous
  - Transformed variables (like logs, see Ch. 7 and will discuss next week)
- Discrete
  - Dummy (AKA binary or dichotomous) variables
  - Ordinal variables
  - Categorical (aka Nominal) variables

- Data sets themselves also come in different forms:
  - Cross-sectional data
  - Panel (time series) data

Binary Independent Variables aka

Dummy Variables

Binary variables

- Dummy variable - a variable with two distinct categories
- Either on or off, yes/no, war/peace etc.
- take a value of 1(on) or 0 (off)
- Dichotomous, binary, dummy - same thing

Binary variables

- Example: They are used in experiments to identify the treated (1) and control (0) units.
- Other Examples:
- This week we will only focus on dummy variables as the **Independent** variable

Using Dummy Variables in Regression

- A dummy variable is equivalant to a difference in means between the two groups.

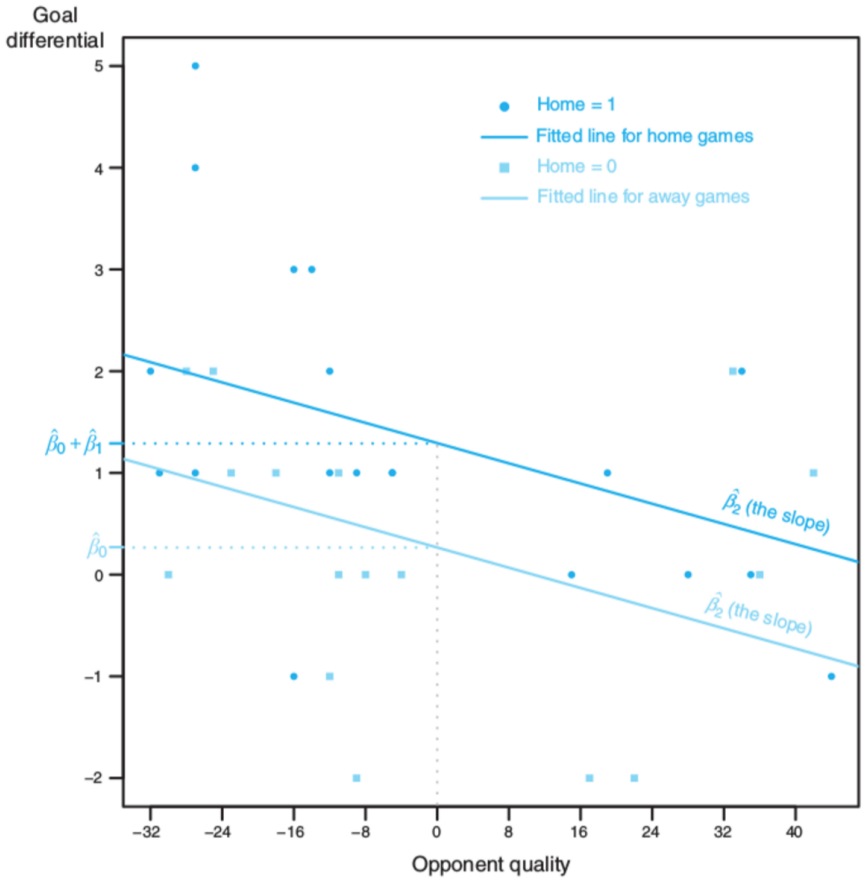


FIGURE 6.7: Fitted Values for Model with Dummy Variable and Control Variable: Manchester City Example

- Fits into our model like any other variable

$$Y_i = \beta_0 + \beta_1 Dummy_i + \epsilon_i$$

- $\beta_1 X_i$ is equal to the difference in means between the two groups
- β_0 is equal to the mean when dummy = 0
- If bivariate model, can plug into equation to get the mean for each group
- statistical significance gives us same p-value as t-test

Dummy Variables and Experiments

- In experiments, binary variables make difference in means across two groups, or ‘average treatment effects’, very easy to calculate.
- Average Treatment effect (ATE) - The ATE measures the difference in mean (average) outcomes between units assigned to the treatment and units assigned to the control.
- i.e. ”on average, this was the effect of the treatment"
- interpret: sign, size, and substantive
- If properly randomized etc. and statistically signficant, this is causal(We’ll talk about this in a couple of weeks)

Dummy Variables in Multivariate OLS

- Back to our Model

$$Y_i = \beta_0 + \beta_1 Dummy_i + \beta_2 X_i + \epsilon_i$$

- Interpretation: Difference in groups, controlling for . . . . . .


weight (tons) = 2.4 + 0.6*(spotted) + ...
pattern reference level: Striped
if other variables constant, spotted dragons will weigh 0.6 tons more than striped dragons, on average.
@allison_horst

Categorical Independent Variables

Categorical Independent Variables

Like binary IVs, but different

Again: Discrete vs Continuous Data

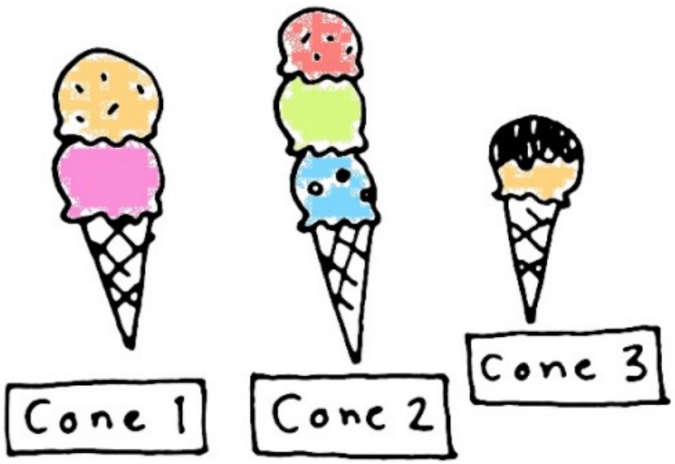
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Again: Discrete vs Continuous Data

- Discrete data: Comes in 'bins' or groups. Example:
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Again: Discrete vs Continuous Data

- Discrete data: Comes in 'bins' or groups. Example:
- Continuous data: Can take any value in a sequence. Examples: Annual income, votes for each candidate, percentages. Note: you can turn continuous data into discrete data. How?



Discrete data:
- There are 3 cones
- Cone 1 has 2 scoops

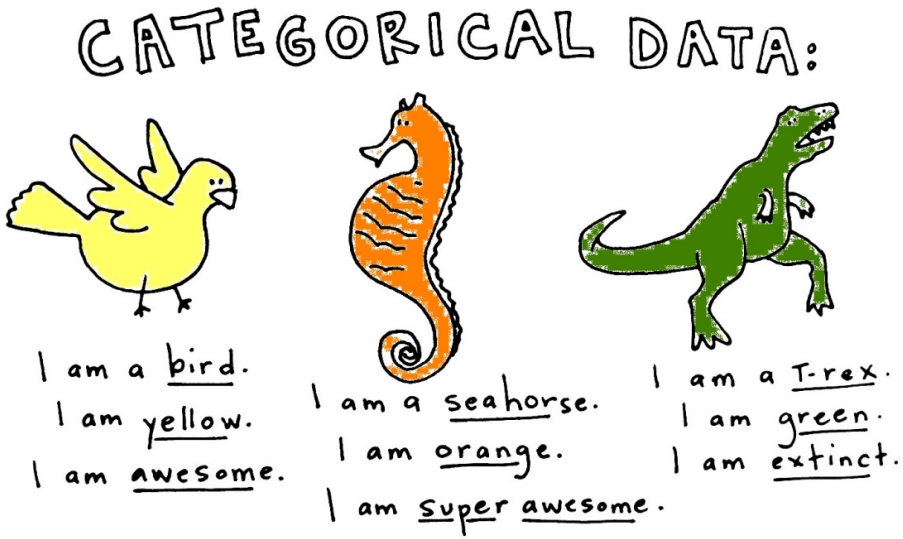
Continuous data:
- Cone 3 weighs 79.4 grams
- cone 2 ice cream is at 8.3°F

Categorical Data

- It is descriptive: it describes how the world is. Often, categorical data comes from qualitative research

Categorical Data

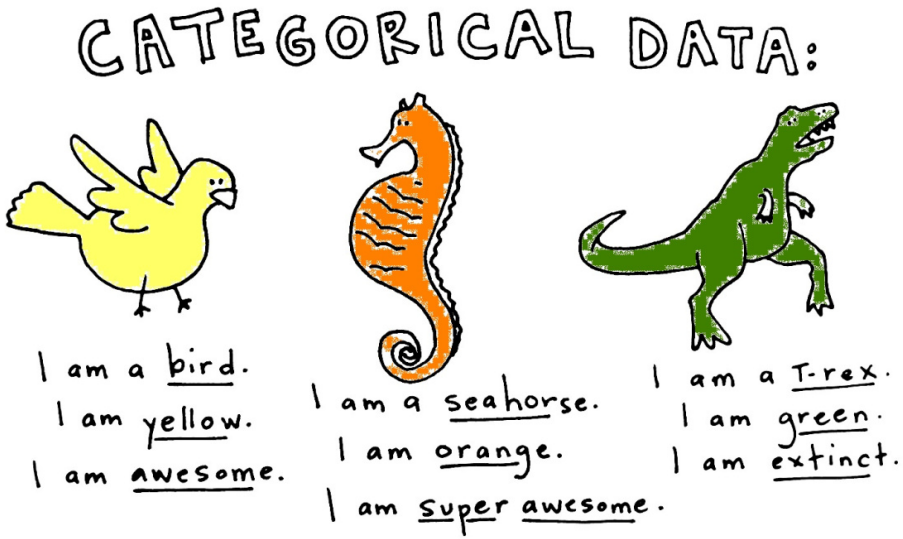
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Categorical Data

- It is descriptive: it describes how the world is. Often, categorical data comes from qualitative research



- Categorical data can be ordinal or nominal. What does this mean? Ordinal can be ordered (low, medium high) versus nominal cannot be ordered (majors: political science, economics, sociology).

Transforming Categorical Data into Discrete Data R

- We might want to turn a category into discrete or binary data, like treated versus control.
- In R we often call categorical data a 'factor'.
- To analyze categorical data we convert each category to dummy variable

  - Some useful commands in R using factors, for example to allow you to translate a categorical variable into a binary (dummy) variable:

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    - lev <- levels(data$factor) # to find out the levels of that factor and store it as an object
    - If not careful R will treat these levels as a continuous variable

Including Categorical Variables in your Regression

- We add categories to our regression model, but leave out our reference category"

$$Y_i = \beta_0 + \beta_1 Category1_i + \beta_2 Category2_i + \epsilon_i$$

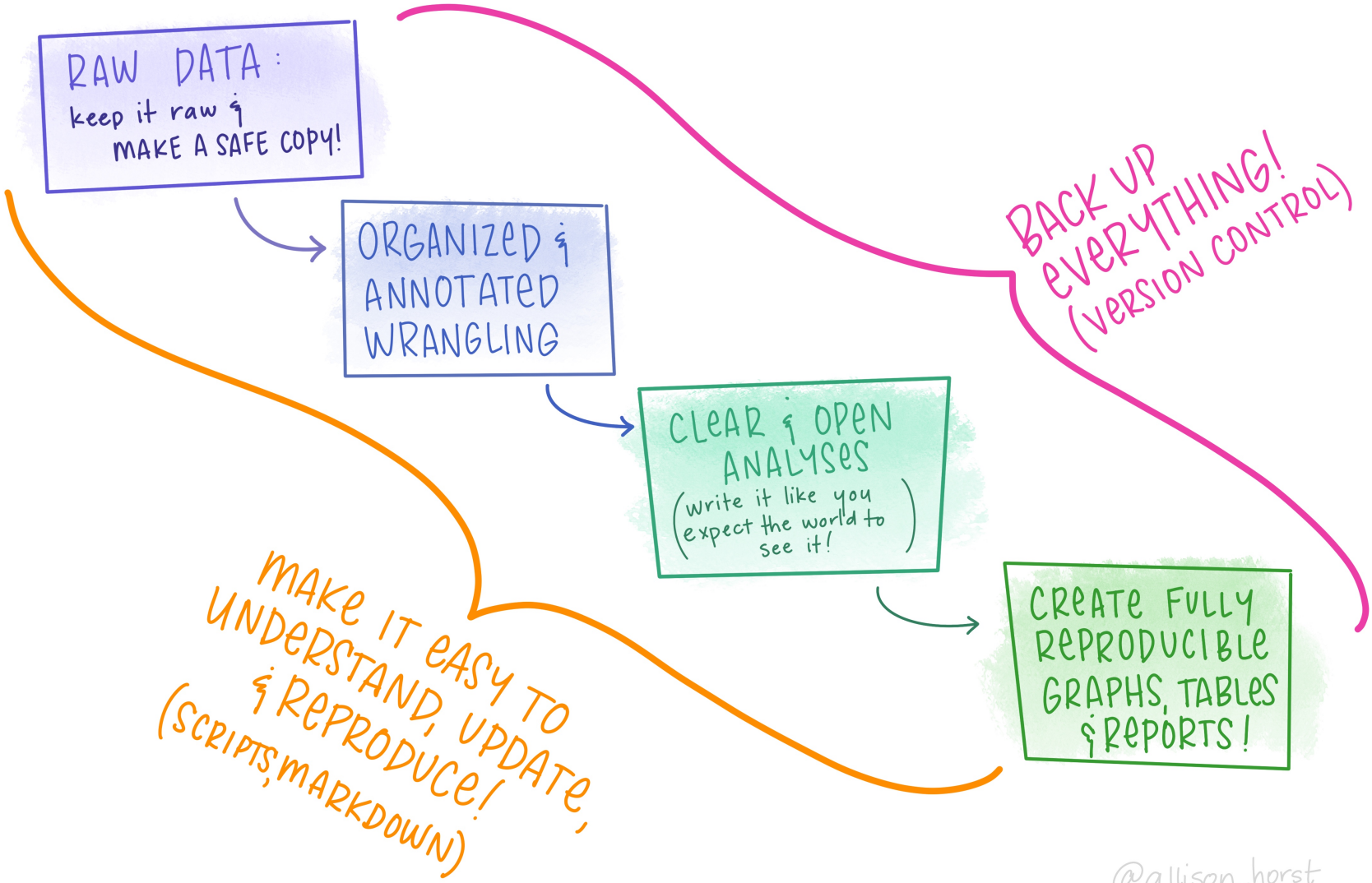
(Note: Depending on data structure in R this could be one factor variable or multiple dummy variables)

- Interpretation of p-value is the same
- Size of coefficient is interpreted as the difference in dependent between that category and the reference category
- Like a dummy, this has the effect of shifting the Y-intercept

Key Terms

- Dummy Variable
- Binary/Dichotomous
- Average Treatment Effect(ATE)
- Categorical
- Nominal
- Ordinal
- Reference Category

- Next Week: Regression Assumptions & Transformations
- Bailey Ch. 7 and Reread Bailey 150-158


RAW DATA:
keep it raw &
MAKE A SAFE COPY!
ORGANIZED & ANNOTATED WRANGLING
CLEAR & OPEN ANALYSES
(write it like you expect the world to see it!)
CREATE FULLY REPRODUCIBLE GRAPHS, TABLES & REPORTS!
BACK UP EVERYTHING! (VERSION CONTROL)
MAKE IT EASY TO UNDERSTAND, UPDATE, & REPRODUCE! (SCRIPTS, MARKDOWN)
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